

Приложение А

Результаты параметризации

В таблице А.1:

- max – максимальное отклонение (сотни км);
- med – медианное отклонение (сотни км);
- avg – среднее отклонение (сотни км).

Таблица А.1 — Результаты параметризации муравьиного алгоритма (начало)

Параметры			Граф 1			Граф 2			Граф 3		
α	p	Дни	max	med	avg	max	med	avg	max	med	avg
0.1	0.1	100	7.5	1.9	2.4	2.8	0.4	0.6	43.5	25.9	26.4
0.1	0.1	200	2.4	0.4	0.9	2.0	0.0	0.4	36.5	22.2	18.1
0.1	0.1	300	1.9	0.0	0.3	0.8	0.0	0.2	37.0	17.0	19.7
0.1	0.1	400	3.4	0.0	0.5	0.8	0.0	0.1	30.0	16.6	15.5
0.1	0.1	500	0.4	0.0	0.1	0.8	0.0	0.1	24.0	16.6	12.3
0.1	0.25	100	7.1	1.9	2.6	7.8	0.8	1.6	40.0	29.9	30.3
0.1	0.25	200	3.0	1.9	1.6	2.0	0.0	0.4	30.0	19.0	16.0
0.1	0.25	300	2.4	0.2	0.6	2.0	0.0	0.6	33.0	18.5	17.3
0.1	0.25	400	3.4	0.0	0.9	0.8	0.0	0.2	28.0	5.0	10.9
0.1	0.25	500	3.0	0.2	0.8	0.0	0.0	0.0	26.0	16.6	13.8
0.1	0.5	100	7.0	2.2	2.8	4.0	1.4	1.6	39.2	26.5	26.6
0.1	0.5	200	4.4	1.9	1.9	0.8	0.0	0.2	31.0	18.6	17.0
0.1	0.5	300	2.4	0.2	0.9	2.0	0.0	0.3	31.0	16.6	18.3
0.1	0.5	400	0.4	0.0	0.2	0.0	0.0	0.0	24.0	15.6	11.5
0.1	0.5	500	0.4	0.2	0.2	0.0	0.0	0.0	24.7	10.6	11.1
0.1	0.75	100	6.4	1.9	2.2	5.8	0.0	1.1	34.0	20.0	19.9
0.1	0.75	200	2.4	1.9	1.3	2.0	0.4	0.6	32.0	21.1	15.9
0.1	0.75	300	2.4	1.1	1.1	0.8	0.0	0.2	24.0	20.5	19.0
0.1	0.75	400	2.4	0.4	0.9	0.8	0.0	0.1	26.0	15.6	11.1
0.1	0.75	500	3.0	0.2	0.6	0.8	0.0	0.1	21.0	10.0	8.8
0.1	0.9	100	4.4	2.0	1.9	2.8	0.4	0.7	37.0	29.0	24.7

Таблица А.1 — продолжение

Параметры			Граф 1			Граф 2			Граф 3		
α	p	Дни	max	med	avg	max	med	avg	max	med	avg
0.1	0.9	200	4.4	0.4	1.3	2.8	0.0	0.5	34.0	18.5	19.3
0.1	0.9	300	2.0	0.4	0.4	0.8	0.0	0.1	35.0	10.6	12.4
0.1	0.9	400	2.0	0.0	0.6	0.0	0.0	0.0	20.0	10.0	8.9
0.1	0.9	500	2.0	0.0	0.2	0.0	0.0	0.0	20.0	5.0	7.9
0.25	0.1	100	9.0	2.0	2.7	2.8	1.4	1.3	43.5	25.5	26.9
0.25	0.1	200	5.0	1.9	2.0	0.8	0.0	0.3	35.0	24.0	22.9
0.25	0.1	300	4.9	0.9	1.5	0.8	0.0	0.2	30.0	24.0	21.9
0.25	0.1	400	1.9	0.4	0.7	0.0	0.0	0.0	22.0	17.0	13.7
0.25	0.1	500	2.4	1.1	1.1	0.0	0.0	0.0	17.0	5.0	7.4
0.25	0.25	100	7.0	3.6	3.2	4.0	0.4	1.2	43.0	25.0	25.1
0.25	0.25	200	3.0	1.1	1.4	2.8	0.0	0.3	33.0	20.0	17.8
0.25	0.25	300	3.9	0.4	1.4	2.0	0.0	0.4	27.2	20.5	17.5
0.25	0.25	400	1.9	0.4	0.4	0.8	0.0	0.1	36.0	11.0	13.6
0.25	0.25	500	0.4	0.0	0.2	0.0	0.0	0.0	25.0	10.0	9.4
0.25	0.5	100	5.7	2.0	2.8	4.3	0.8	1.4	42.0	36.0	32.3
0.25	0.5	200	3.0	1.1	1.2	0.8	0.0	0.1	34.0	19.5	16.8
0.25	0.5	300	3.0	0.4	0.7	0.8	0.0	0.2	32.0	16.0	15.0
0.25	0.5	400	2.4	1.1	1.3	0.0	0.0	0.0	26.0	18.1	14.5
0.25	0.5	500	1.9	0.2	0.3	0.0	0.0	0.0	26.0	16.0	13.8
0.25	0.75	100	5.0	2.2	2.4	5.8	0.0	1.3	37.0	29.6	27.7
0.25	0.75	200	3.0	1.2	1.4	0.8	0.0	0.3	37.0	19.0	16.3
0.25	0.75	300	3.0	0.4	0.8	0.8	0.0	0.1	35.5	24.8	24.0
0.25	0.75	400	2.0	0.4	0.4	0.8	0.0	0.1	28.0	17.5	15.3
0.25	0.75	500	2.0	0.0	0.5	2.0	0.0	0.2	21.2	5.0	9.0
0.25	0.9	100	3.0	1.1	1.4	2.8	0.8	1.0	40.0	23.8	20.2
0.25	0.9	200	4.4	1.2	1.7	0.8	0.0	0.2	26.0	7.5	9.4
0.25	0.9	300	2.0	0.9	1.0	2.0	0.0	0.2	22.0	5.0	8.0
0.25	0.9	400	2.0	0.2	0.4	0.8	0.0	0.1	25.0	18.1	13.4
0.25	0.9	500	2.0	0.4	0.7	0.0	0.0	0.0	17.0	0.0	4.2

Таблица А.1 — продолжение

Параметры			Граф 1			Граф 2			Граф 3		
α	p	Дни	max	med	avg	max	med	avg	max	med	avg
0.5	0.1	100	10.9	6.2	5.6	6.6	2.0	2.5	46.0	38.0	32.6
0.5	0.1	200	3.4	1.1	1.3	4.0	1.4	1.6	39.0	23.0	22.0
0.5	0.1	300	6.0	1.2	2.3	2.0	0.0	0.4	32.0	21.0	16.7
0.5	0.1	400	2.4	0.4	0.8	2.0	0.0	0.3	30.0	16.6	15.5
0.5	0.1	500	3.0	1.9	1.4	0.8	0.0	0.2	21.0	10.0	9.6
0.5	0.25	100	6.4	3.4	3.2	7.8	2.0	2.4	44.0	29.6	26.0
0.5	0.25	200	5.7	2.0	2.6	2.0	0.0	0.4	44.0	29.0	26.2
0.5	0.25	300	6.4	0.4	1.8	2.0	0.4	0.6	26.0	15.6	13.4
0.5	0.25	400	3.0	0.4	0.7	2.0	0.0	0.2	26.0	18.1	14.5
0.5	0.25	500	3.4	0.4	0.9	0.0	0.0	0.0	27.0	19.0	18.3
0.5	0.5	100	8.0	4.7	4.3	7.8	0.8	1.9	36.7	22.6	21.0
0.5	0.5	200	4.4	2.5	1.9	2.8	0.0	0.9	36.0	19.2	16.4
0.5	0.5	300	2.0	0.2	0.7	0.8	0.0	0.2	36.0	17.0	16.0
0.5	0.5	400	1.9	0.2	0.3	0.8	0.0	0.2	31.0	16.0	12.9
0.5	0.5	500	5.7	0.4	1.3	0.8	0.0	0.2	26.0	16.2	11.8
0.5	0.75	100	7.4	1.2	2.9	2.8	1.4	1.2	43.0	24.0	21.7
0.5	0.75	200	6.0	1.7	2.5	6.8	0.4	1.0	23.5	17.0	12.8
0.5	0.75	300	4.4	0.4	0.9	0.8	0.0	0.2	26.0	16.0	14.6
0.5	0.75	400	3.4	0.4	0.8	0.0	0.0	0.0	31.2	15.6	15.1
0.5	0.75	500	0.4	0.2	0.2	0.8	0.0	0.2	22.0	5.0	7.5
0.5	0.9	100	6.4	2.5	2.6	5.8	0.4	1.1	43.0	18.5	21.7
0.5	0.9	200	7.0	2.0	2.3	4.3	0.0	0.9	27.0	19.1	17.2
0.5	0.9	300	4.4	0.4	1.2	0.0	0.0	0.0	21.2	2.5	7.6
0.5	0.9	400	3.0	0.4	0.7	0.0	0.0	0.0	17.0	5.0	5.3
0.5	0.9	500	2.0	0.4	0.4	0.8	0.0	0.1	20.0	10.0	9.4
0.75	0.1	100	9.4	5.0	5.4	12.1	3.4	4.6	57.5	44.0	40.8
0.75	0.1	200	9.0	4.4	4.4	6.8	3.5	3.2	50.0	33.0	27.6
0.75	0.1	300	8.1	2.0	3.0	7.8	0.8	1.7	38.0	28.0	26.4
0.75	0.1	400	6.0	1.9	2.4	2.8	0.0	0.4	37.0	18.5	19.8

Таблица А.1 — продолжение

Параметры			Граф 1			Граф 2			Граф 3		
α	p	Дни	max	med	avg	max	med	avg	max	med	avg
0.75	0.1	500	3.9	0.4	1.4	7.8	0.0	1.1	34.0	23.5	21.6
0.75	0.25	100	15.4	6.0	6.3	13.8	4.3	5.6	50.0	31.0	32.3
0.75	0.25	200	8.0	1.9	3.2	6.5	1.4	2.0	47.0	19.5	23.7
0.75	0.25	300	8.0	2.9	3.4	9.8	1.4	2.8	35.0	25.9	22.9
0.75	0.25	400	4.4	1.9	2.0	6.6	0.0	1.0	30.0	18.5	15.6
0.75	0.25	500	4.4	2.0	2.1	6.5	0.8	1.6	35.0	17.0	17.2
0.75	0.5	100	12.9	4.2	4.8	14.7	2.0	3.8	49.2	30.8	30.7
0.75	0.5	200	6.4	3.2	3.4	7.8	0.4	1.8	38.7	29.0	27.4
0.75	0.5	300	7.5	2.9	3.2	2.8	0.8	1.0	30.0	16.0	13.5
0.75	0.5	400	4.4	2.4	2.3	5.8	0.4	1.9	32.0	22.4	17.9
0.75	0.5	500	4.4	2.0	2.1	2.8	0.0	0.6	30.0	10.6	12.9
0.75	0.75	100	9.0	4.2	4.5	16.0	2.0	3.6	46.5	27.5	24.2
0.75	0.75	200	4.9	2.9	2.7	10.1	2.8	4.0	45.0	26.5	22.8
0.75	0.75	300	5.0	2.0	2.1	4.0	1.4	1.6	30.0	22.9	19.5
0.75	0.75	400	6.4	3.4	2.9	2.0	0.4	0.5	32.0	18.5	17.1
0.75	0.75	500	3.9	0.4	0.9	2.0	0.0	0.4	32.0	19.0	15.2
0.75	0.9	100	7.7	2.4	4.0	13.3	4.3	5.5	39.0	22.4	21.6
0.75	0.9	200	8.4	4.7	4.0	4.3	0.8	1.2	34.0	23.0	19.1
0.75	0.9	300	4.9	2.0	1.9	2.0	0.4	0.5	26.0	15.6	14.3
0.75	0.9	400	5.7	0.4	1.4	4.0	0.0	0.7	32.0	21.0	17.9
0.75	0.9	500	2.4	0.4	0.6	0.8	0.0	0.2	32.0	10.6	10.9
0.9	0.1	100	18.7	9.7	9.8	20.6	11.8	10.4	69.0	51.1	48.4
0.9	0.1	200	15.9	6.4	8.0	13.3	5.8	6.5	43.0	29.0	29.6
0.9	0.1	300	7.7	3.5	3.5	12.0	5.0	5.0	46.5	34.0	30.2
0.9	0.1	400	7.0	4.7	4.3	11.3	2.0	3.2	42.0	34.2	32.2
0.9	0.1	500	9.9	2.2	3.4	7.8	0.4	2.6	47.0	24.6	26.1
0.9	0.25	100	17.4	11.8	11.1	19.9	13.9	12.3	56.0	41.8	41.3
0.9	0.25	200	7.6	5.5	4.9	12.6	3.7	5.0	50.0	38.2	35.2
0.9	0.25	300	12.4	5.3	5.4	12.3	5.2	5.7	45.0	31.1	27.6

Таблица А.1 — продолжение

Параметры			Граф 1			Граф 2			Граф 3		
α	p	Дни	max	med	avg	max	med	avg	max	med	avg
0.9	0.25	400	9.4	5.2	5.1	13.3	2.4	3.5	39.0	30.5	27.9
0.9	0.25	500	6.4	2.2	3.3	12.6	2.0	4.0	31.0	25.0	22.4
0.9	0.5	100	11.9	8.2	8.2	17.8	10.4	9.3	55.7	43.9	43.7
0.9	0.5	200	9.0	5.3	5.4	19.6	8.0	8.0	54.2	33.5	34.6
0.9	0.5	300	12.4	3.1	4.2	13.3	1.8	3.4	39.0	30.0	28.5
0.9	0.5	400	7.1	4.7	4.3	12.3	3.0	3.3	46.0	26.5	22.2
0.9	0.5	500	9.0	4.7	4.3	6.6	0.4	1.5	43.0	26.0	24.3
0.9	0.75	100	14.6	9.2	9.0	23.1	12.6	11.3	47.2	41.5	38.4
0.9	0.75	200	13.1	7.0	7.7	6.6	3.5	3.2	48.0	30.5	30.0
0.9	0.75	300	10.1	4.7	4.6	15.4	1.4	3.6	41.2	30.0	27.5
0.9	0.75	400	10.1	3.4	4.3	13.3	3.0	4.8	42.0	26.5	25.7
0.9	0.75	500	9.0	3.4	3.6	4.3	2.4	2.1	37.0	20.6	18.3
0.9	0.9	100	15.4	9.5	9.7	17.6	7.3	8.8	59.2	46.0	44.2
0.9	0.9	200	12.0	6.0	6.1	14.1	2.0	3.5	56.2	31.0	32.3
0.9	0.9	300	8.0	2.7	3.3	10.1	2.0	3.5	40.5	34.2	31.7
0.9	0.9	400	8.4	3.4	3.6	5.8	2.0	1.8	44.2	23.1	21.2
0.9	0.9	500	7.9	4.4	4.0	2.0	0.0	0.6	37.0	11.0	15.6